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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF LAND SURVEYING AND GEO-INFORMATICS

Programme	**:**	34478-SY BSc (Hons) Land Surveying & Geo-Informatics 34480-FYL BSc (Hons) Land Surveying & Geo-Informatics 34480-FYU BSc (Hons) Urban Informatics & Smart Cities 34480-SYL BSc (Hons) Land Surveying & Geo-Informatics 34480-XAL BSc (Hons) Land Surveying & Geo-Informatics with a Secondary Major In Aida 62428-DSA BSc (Hons) Data Science & Analytics 62428-SYD BSc (Hons) Data Science & Analytics			
Subject Title	**:**	Intelligent Transportation Systems			
Subject Code	**:**	LSGI3213	**Session**	**:**	Semester 1, 2025/26
Date	**:**	15 December 2025	**Time**	**:**	12:30 – 14:30
Time Allowed	**:**	2 hours	**Subject Examiner(s)**	**:**	Prof Chengxiang ZHUGE (LSGI)

This question paper has a total of 4 pages.

Instructions to Candidates:

This paper has **TWO** sections, A and B.

Section A: Answer **ALL THREE** questions.
Section B: Answer any **TWO** questions.

All questions carry equal marks
Total marks = 100.

Available from Invigilator: Nil

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Section A: Answer **ALL THREE** questions.

Question 1

Describe the key differences between Logistics and Freight Transport. (20 marks)

Question 2

Answer the following two questions about the travel demand characteristics.

(a) Why is travel demand (or transport) generally considered as a derived demand? (10 marks)

(b) Why is travel demand closely associated with land use? (10 marks)

Question 3

Shortest path algorithms are a family of algorithms designed to solve the shortest path problem and have been widely used in many Intelligent Transportation Systems (ITS). Dijkstra's algorithm is a typical solution to the single-source shortest path problem in graph theory. As shown in Figure 1, the directed graph is composed of five nodes/vertices (i.e., A, B, C, D, and E) and several directed links/edges. The numbers next to the links indicate the weights (also known as travel costs in transportation studies). Figure out the shortest paths from Node A to the rest of the Nodes B, C, D and E using the Dijkstra's algorithm, and calculate the total weight/travel cost for each of the four shortest paths.

(a) Write down the shortest path from Node A to Node B (A→B) using a sequence of visited nodes (including start and end nodes), and its weight/travel cost. (5 marks)

(b) Write down the shortest path from Node A to Node C (A→C) using a sequence of visited nodes (including start and end nodes), and its weight/travel cost. (5 marks)

(c) Write down the shortest path from Node A to Node D (A→D) using a sequence of visited nodes (including start and end nodes), and its weight/travel cost. (5 marks)

(d) Write down the shortest path from Node A to Node E (A→E) using a sequence of visited nodes (including start and end nodes), and its weight/travel cost. (5 marks)

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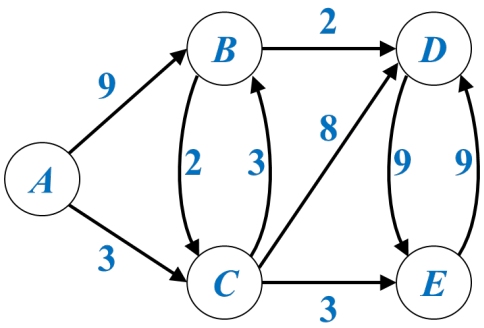


Figure 1 A directed graph for the Dijkstra's algorithm.

- END OF SECTION A -

Section B: Answer any **TWO** questions.

Question 4

The Hong Kong government has put efforts into electrification of transportation, and the number of electric vehicles is on the rise. Describe two specific Intelligent Transportation Systems (ITS) that can be developed for electric vehicle drivers in Hong Kong when the drivers want to park or charge their electric vehicles at their trip destinations. Your description should include:

(a) What are the key functions of the ITS that you choose? (10 marks)

(b) What key technologies or datasets are needed for the development of the ITS that you choose? (10 marks)

Question 5

Bike sharing is a typical shared mobility service. Describe two specific Intelligent Transportation Systems (ITS) that can be developed for shared bike users in Hong Kong. Your description should include:

(a) What are the key functions of the ITS that you choose? (10 marks)

(b) What key technologies or datasets are needed for the development of the ITS that you choose? (10 marks)

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Question 6

Vehicle trajectory data is a typical spatial data type. Describe two specific Intelligent Transportation Systems (ITS) that can be developed for Hong Kong, using vehicle trajectory data. Your description should include:

(a) What are the key functions of the ITS that you choose? (10 marks)

(b) How can the vehicle trajectory data be used to develop the ITS that you choose? (10 marks)

- END OF SECTION B -

- END OF PAPER -